

NORTHEASTERN UNIVERSITY
MIE Department
IE5390 Excel

FINAL PROJECT

Assigned 2/28/25

Due: 4/9/25 by 11:59 PM. 50% till 4/11 by 11:59 PM. **ZERO afterwards. NO Exceptions**

Final Project: Data Analysis using R and Python

NOTES:

- **You can mix using R and Python**
- **decide which project tasks are better using R or Python**

Task:

Perform an original analysis on a dataset of your choosing. The dataset can come from any source as long as it is something you have not already worked on for this course and meets the requirements as follows:

- The dimensions of the dataset consist of at least 5000 rows (records) and 8-10 columns (variables)
- There are sufficient missing values in the data – enough to demonstrate cleaning/manipulation
- The data selected makes logical sense and are not just randomly generated values

Possible sources of data include Data sets from:

- UCI Machine Learning Repository: <https://archive.ics.uci.edu/ml/datasets.php>
- Kaggle datasets: <https://www.kaggle.com/datasets>
- FiveThirtyEight: <https://data.fivethirtyeight.com/>
- Data Gov: <https://www.data.gov/>

Note that these are the recommended repositories that provide data collected from legitimate studies with reliable sources. You may use data from other sources if they meet the requirements stated above.

Your analysis should clearly demonstrate the tools and techniques you've been exposed to in this course.

This can take any form you'd like (i.e., comparison of averages across categories, mapping data, developing reports/data tables and visualizing Key-Performance-Indicators). You **MUST** use R and Python to do some sort of analysis.

Deliverables:

1. A report that documents your data set selection, activities you did and why you did them with enough screenshots to show steps, code and results. Basically, think of yourself as a data analyst, providing a insight and analysis to upper management to uncover hidden trends.
 - a. Describe the scenario. What question do you want to answer and why is it important?
 - b. describe the data. What are the key elements
 - c. Data set file you used and the url where you found it
 - d. Describe the characteristics of the data in an 'as-is' form. Were there any missing values? Any weird anomalies? Make good use of data visualizations **if needed**
 - e. Discuss the analysis conducted and the results. **If needed**, use infographic to walk-through the process that you followed, i.e. flowchart, decision-tree, etc.
 - f. Discuss why you used R or Python for tasks where you used them

- g. Describe your conclusions. What did you learn? Support your conclusions using the results of the analysis, citing specific evidence!
2. Submit both the R files and Python files you have created, clearly labeled and annotated with comments inside the files
3. List of references and resources you used to complete the project
4. Any other material you need to include

Code Requirements:

The code should be written using the following format:

- The code is written in either R Markdown (Notebook) and in Jupyter Notebook format
- Thoroughly labelled section headings and descriptors with explanations for each section detailing the tasks undertaken along
- Comments provided for the code that briefly describe the functions/methods used and the objective of the code chunk/cell

Final Project - Guidelines:

You must work individually for this project. You may discuss and brainstorm ideas with other students from the class to obtain peer-feedback, but **you must work on your project by yourself**.

Final Project - What to Submit and Where:

Submit the following documents as attachments via Canvas and follow the naming convention:

1. Report named as **LastName_report (word doc or PDF)**
2. data set named as **LastName_dataset.xlsx (or csv format)**
3. Code in PDF format as **LastName_Code.pdf**
4. Code in native R or Python format as **LastName_Code.rmd** or **LastName_Code.ipynb**